

MATH3026/4022 Lecture 2

Stationarity and Key Operators

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Outline

- ▶ Stationarity and its importance in time series analysis.
- ▶ Definitions of key features of time series:
 - ▶ Mean function
 - ▶ Autocovariance function
 - ▶ Autocorrelation function
- ▶ Key operators used in time series analysis:
 - ▶ Backward shift and forward shift operators
 - ▶ Difference and sum operators
- ▶ Transforming non-stationary time series to stationary time series.

Time Series: Recap

In Lecture 1 we saw that a **time series** occurs where we have a **sequence of measurements** made **over time**.

When summarising time series data we tend not to use methods encountered in previous courses.

This is because of the time-ordered and correlated nature of time series data.

When modelling a time series, we must consider the **variation** and how the **variation may change over time**.

Typically, the variation that we might seek to model consists of **seasonality** and **trend**.

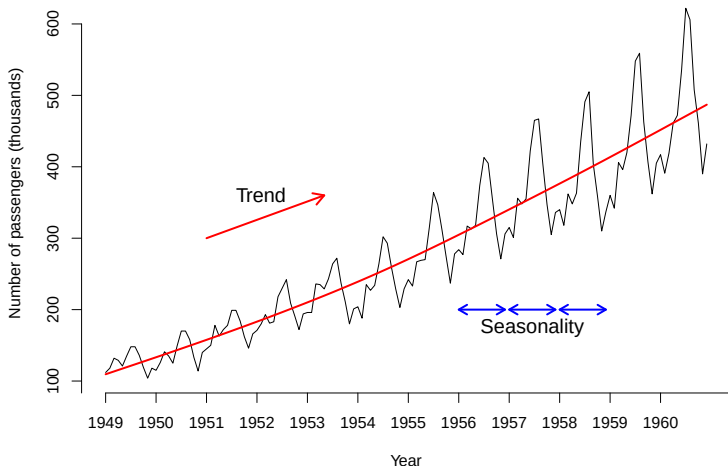
Recapping from Lecture 1:

Seasonality refers to how observations may fluctuate or change in a short, cyclic period.

Trend refers to how data may change in general in the long term.

Modelling time series

For example, the time plot below shows monthly US air passenger numbers for a US airline for the years 1949 to 1960.



Modelling time series

We see that there is a trend where the number of passengers has increased, in general, during the years 1949 to 1960.

In addition, there is obvious **seasonality** in the data.

Within each calendar year the number of passengers tends to be low in January then increases through to a peak during July/August before decreasing again into the autumn/winter months at the end of the year.

Also, we note that the **variation** in the data is increasing as time increases.

We can see this by observing how the distance between the highest and lowest monthly passenger numbers gets larger as the year increases.

Stationarity

Despite our obvious interest in features such as **trend** and **seasonality**, when modelling time series we often seek to model what is known as a **stationary** time series.

This is for mathematical convenience and will become apparent later in the course.

First, we shall describe in words (and formally) what is meant by a **stationary time series**.

A time series is said to be **stationary** if:

- ▶ It contains no trend (i.e. no systematic change in mean over time).
- ▶ It contains no systematic change in variance over time.
- ▶ Strictly periodic variations (i.e. seasonality) have been removed.

Stationarity

In essence, if a time series is **stationary** then **one section of the data** will look **similar** to any **other section of the data**.

We shall now introduce a more formal, mathematical, definition of stationarity.

Definition 2.1: Strict Stationarity

Let $\{X_t : t \in \mathbb{Z}\}$ be a time series or process.

The time series (process) is said to be **strictly stationary** if the joint distribution of $X_{t_1}, X_{t_2}, \dots, X_{t_k}$ is the same as the joint distribution of $X_{t_1+\tau}, X_{t_2+\tau}, \dots, X_{t_k+\tau}$ for all $t_1, \dots, t_k$ and τ , where $\tau \in \mathbb{Z}$.

In practice, strict stationarity is fairly restrictive.

Indeed, this property would imply that all finite dimensional distributions of the stochastic process $\{X_t\}$ are identical – the probabilistic structure would not change over time.

Instead, we often make use of a property known as **weak stationarity** or **second-order stationarity** when modelling time series.

Mean and Covariance Function

Suppose $\{X_t : t \in \mathbb{Z}\}$ is a process with $\mathbb{E}(X_t^2) < \infty$ for all $t \in \mathbb{Z}$.

Definition 2.2: Mean Function and Covariance Function

The **mean function** of $\{X_t : t \in \mathbb{Z}\}$ is defined

$$\mu(t) = \mathbb{E}(X_t).$$

The **covariance function** of $\{X_t : t \in \mathbb{Z}\}$ is defined

$$\begin{aligned}\gamma(r, s) &= \text{Cov}(X_r, X_s) \\ &= \mathbb{E}[(X_r - \mu(r))(X_s - \mu(s))]\end{aligned}$$

for all $(r, s) \in \mathbb{Z} \times \mathbb{Z}$.

Definition 2.3: Weak Stationarity

The time series $\{X_t : t \in \mathbb{Z}\}$ is said to be **weakly stationary** or **second-order stationary** if

- (i) $\mu(t)$ is independent of t .
- (ii) $\text{Var}(X_t) = \sigma^2(t) = \sigma^2 < \infty$ for all t .
- (iii) $\gamma(t, t + h)$ is independent of t for each $h \in \mathbb{Z}$.

Stationarity

Considering each of the above points in words, we note that, for a weakly stationary process:

- (i) $\mu(t)$ is independent of t implies that $\mu(t) = \mu$ is constant for all t – the mean of the time series does not change over time.
- (ii) The time series has a constant variance at each time point.
- (iii) The covariance between X_t and X_{t+h} depends only on the distance in time between points $|h|$ and not on the value of t .

Owing to point (iii), we see that, for a weakly stationary time series, we can define

$$\gamma(h) = \gamma(0, h) = \gamma(t, t + h) \text{ for all } t.$$

We call $\gamma(\cdot)$ the **autocovariance** function and $\gamma(h)$ the value of the autocovariance function at lag h .

Autocorrelation Function

Linked to the autocovariance function is the **autocorrelation function**, which we define below.

Definition 2.4: Autocorrelation Function

The **autocorrelation function** of the time series $\{X_t : t \in \mathbb{Z}\}$ at lag h is defined

$$\begin{aligned}\rho(h) &= \text{Cor}(X_t, X_{t+h}) \\ &= \frac{\text{Cov}(X_t, X_{t+h})}{\sqrt{\text{Var}(X_t)\text{Var}(X_{t+h})}} \\ &= \frac{\gamma(h)}{\gamma(0)}.\end{aligned}$$

In MATH3026/4022, whenever the term **stationarity** (or **stationary**) is used with respect to a time series this shall mean **weak stationarity** (or **weakly stationary**).

We can see that, in principle, a stationary time series has some properties that would be desirable when modelling, namely:

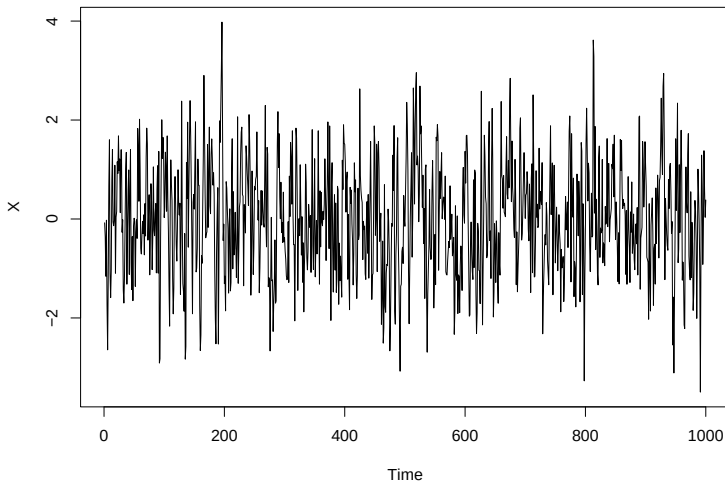
- (i) A constant mean and variance.
- (ii) A time-invariant covariance function.

However, in reality, many time series are not naturally stationary.

We have already seen examples that contain **seasonality** and **trend**.

Stationary Time Series - Example

The plot below is a time plot for a (weakly) stationary time series



Modelling a time series

As a result, when modelling a time series we often seek to:

1. Transform the original time series into a stationary time series.
2. Fit a model to the stationary time series.
3. Use the model from Step 2. to make inference about the original time series.

The steps above are followed because modelling a stationary time series is more straightforward than attempting to directly model a time series where behaviour is changing over time.

The idea is similar to that in linear regression, where we might apply a suitable transformation to data to satisfy modelling assumptions prior to fitting a linear regression model.

Modelling a time series

We note that this is not the *only* method by which a time series may be modelled.

However, for now, we shall concentrate on modelling stationary time series.

To begin with, we define one of the most basic time series, one where the random variables are mutually independent - known as a **purely random process** or **white noise process**.

Purely Random/White Noise Process

Definition 2.5: Purely Random Process/White Noise Process

A stochastic process $\{Z_t : t \in \mathbb{Z}\}$ for which

- (i) The variables $Z_t, t \in \mathbb{Z}$, are **identically distributed** with $\mathbb{E}(Z_t) = 0$ for all t .
- (ii) The variables $Z_t, t \in \mathbb{Z}$, are **independent** ($\gamma_Z(r, s) = 0$ where $r \neq s$).
- (iii) The variables $Z_t, t \in \mathbb{Z}$, have a **constant variance** ($\gamma_Z(t, t) = \sigma_z^2$ where $0 < \sigma_z^2 < \infty$ for all $t \in \mathbb{Z}$).

is known as a **white noise** or **purely random** process.

The white noise process is important when modelling time series.

Purely Random/White Noise Process

Such processes will be helpful when modelling time series as the random variables in a white noise process might be considered as **random error** terms at each time point in a time series (cf. linear modelling).

In words, we might consider modelling a time series (at time t) as

$$g(X_t) = \text{Some function of other variables in the series} + Z_t$$

where Z_t is white noise at time t .

Here $g(\cdot)$ is some function of the time series in which, perhaps, trend and seasonality have been removed.

Key Operators

As noted previously, usually, we seek to make a time series stationary prior to modelling the series.

We shall now introduce some key operators that could be used to do this.

Definition 2.6: Backward Shift Operator

The backward shift operator, B , is defined

$$BX_t = X_{t-1}.$$

Essentially, the backward shift operator, B , shifts a variable **back** by **one unit of time**.

We note that

$$B^2 X_t = B(BX_t) = BX_{t-1} = X_{t-2}$$

and it follows that

$$B^k X_t = X_{t-k} \text{ for all } k \in \mathbb{N}.$$

Definition 2.7: Forward Shift Operator

The forward shift operator, F , is defined

$$F = B^{-1}$$

such that

$$FX_t = X_{t+1}.$$

Essentially, the forward shift operator, F , shifts a variable **forward** by one unit of time

It follows that $F^k X_t = X_{t+k}$ for all $k \in \mathbb{N}$.

Definition 2.8: Backward Difference Operator

The backward difference operator, ∇ , is defined

$$\nabla X_t = X_t - X_{t-1}.$$

It follows that

$$\begin{aligned}\nabla X_t &= X_t - BX_t \\ &= (1 - B)X_t.\end{aligned}$$

Hence, $\nabla = 1 - B$.

Definition 2.9: Summation Operator

The summation operator, $S = \nabla^{-1}$, is defined

$$SX_t = X_t + X_{t-1} + X_{t-2} + \dots$$

We see that

$$SX_t = (1 + B + B^2 + \dots)X_t.$$

Form of the summation operator

Example 2.1

Suppose that the time series $\{X_t : t \in \mathbb{Z}\}$ is defined

$$X_t = X_{t-1} + Z_t$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2 .

- (a) Show that this time series is non-stationary.
- (b) Using a suitable operator, transform this time series into a stationary time series.

Example 2.1

Example 2.1

The time series in Example 2.1 is known as a **random walk**.

A real-world example of a time series that might behave in a similar manner to that in Example 2.1 is a daily share price for a company, whereby

Share price on day t = Share price on day $t - 1$ + Random error.

Example 2.2

The time series $\{X_t : t \in \mathbb{Z}\}$ is defined

$$X_t = \frac{1}{2}X_{t-1} + Z_t$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2 .

Is the process $\{X_t\}$ stationary? Justify your answer.

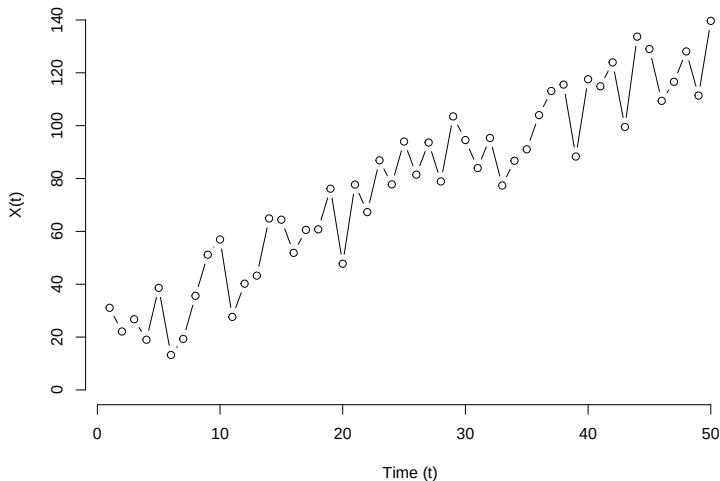
Example 2.2

Example 2.2

Example 2.2

Example 2.3

Consider the time plot below of a time series $\{X_t : t \in \mathbb{Z}^+\}$.



Example 2.3

Examining the time plot, it does not look like the process is stationary.

We observe what appears to be a linear trend (X_t increasing linearly as t increases).

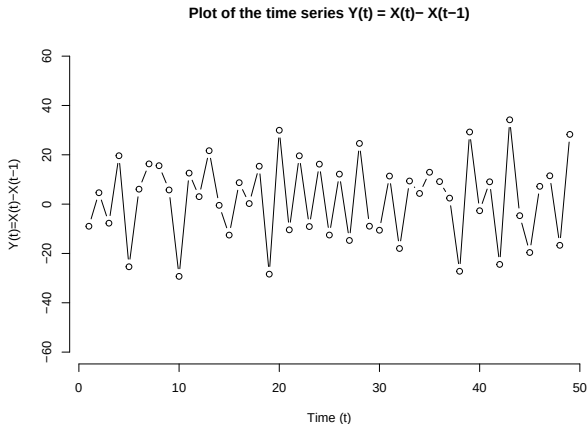
Clearly $\mathbb{E}(X_t)$ does not appear to be constant.

We define the time series $\{Y_t\}$ where

$$Y_t = \nabla X_t = X_t - X_{t-1}$$

We'll show a time plot for the time series $\{Y_t\}$ on the next slide.

Example 2.3



The above (transformed) process now looks stationary and we obtained the stationary process through removal of the linear trend using the **first-order difference**.

Removing trend using the backward shift operator

For a **linear** trend, as we have seen, the first difference might be used to remove trend.

For a **quadratic** trend, we might use second differences where

$$\begin{aligned}Y_t &= (X_t - X_{t-1}) - (X_{t-1} - X_{t-2}) \\&= (1 - B)X_t - (1 - B)X_{t-1} \\&= (1 - B)^2 X_t \\&= \nabla^2 X_t.\end{aligned}$$

In general, if there is systematic trend in the mean that corresponds to a **polynomial of degree** d , then the transformation

$$Y_t = (1 - B)^d X_t$$

can be used to remove this trend.

Removing Trend

We see that one intuitive approach to take when modelling time series data might be as follows:

1. Using observations from the time series, produce a time plot of observed values (x_t) against time (t) and assess whether or not there appears to be any systematic trend (e.g. linear, quadratic etc.).
2. Depending on the time plot from 1., apply different transformations of the form $(1 - B)^d$ to the observed data $\{x_t\}$.
3. Plot the transformed data from 2. against time (t) and assess which of the transformations results in a series that appears stationary (or at least has the trend removed).

In practice, we would often consider only $d = 0, 1, 2$ when applying the above transformation.

Lecture 2: Learning Outcomes

- ▶ Understand what is meant by **stationarity** with regard to a time series and how **strict stationarity** and **weak stationarity** differ.
- ▶ Be able to define and compute the **mean** and **covariance functions** for a time series.
- ▶ Know the definition of and be able to compute the **autocorrelation function** of a time series.
- ▶ Understand what is meant by a **white noise process**.
- ▶ Know and apply several **key operators** used in time series analysis (e.g. backward shift, forward and difference operators).
- ▶ Understand how **key operators** may be used to **transform non-stationary time series to stationary time series**.